

CHAPTER 3

COUNTING

3.1 INTRODUCTION

In the classical interpretation of probability, all outcomes of the experiment are equally likely, and the probability of an event is obtained as the relative frequency of outcomes that favor this event (imply its occurrence). Simple enumeration of elements in these sets is often not feasible, and therefore practical implementation of this principle requires developing techniques for counting elements of certain sets (e.g., sets of all possible outcomes of an experiment). The branch of mathematics dealing with such methods is called *combinatorics*, or *combinatorial analysis*. In this chapter, we introduce some combinatorial principles and illustrate their use in computing probabilities.

A much more complete presentation of combinatorial methods and their applications to probability can be found in Feller (1968).

3.2 PRODUCT SETS, ORDERINGS, AND PERMUTATIONS

Consider two operations of some sort, which can be performed one after another. Leaving the notion of “operation” vague at the moment, we can make two assumptions:

1. The first operation can be performed in k_1 different ways.
2. For each of the ways of performing the first operation, the second operation can be performed in k_2 ways.

We have the following theorem:

Theorem 3.2.1 *Under assumptions 1 and 2, a two-step procedure consisting of a first operation followed by the second operation can be performed in $k_1 k_2$ distinct ways.*

Proof: Observe that each way of performing the two operations can be represented as a pair (a_i, b_{ij}) with $i = 1, \dots, k_1$ and $j = 1, \dots, k_2$, where a_i is the i th way of performing the first operation and b_{ij} is the j th way of performing the second operation if the first operation was performed in i th way. All such pairs can be arranged in a rectangular array with k_1 rows and k_2 columns. $\square$

We will now show some applications of Theorem 3.2.1.

■ EXAMPLE 3.1 Cartesian Products

One of the most common operations on sets is the Cartesian product. If A and B are two sets, their Cartesian product $A \times B$ is defined as the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. For instance, if A consists of elements x and y , while B consists of the digits 1, 2, and 3, then the Cartesian product $A \times B = \{x, y\} \times \{1, 2, 3\}$ contains the six pairs

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}. \quad (3.1)$$

Observe that the Cartesian product $A \times B$ is an operation quite distinct from the set-theoretical product $A \cap B$. For instance, in the above case, $A \cap B = \emptyset$, since A and B have no elements in common. Also, while $A \cap B = B \cap A$, for Cartesian products $A \times B \neq B \times A$ in general. In cases when there is no danger of confusion, we will use the term *product* for *Cartesian product*.

Identifying now the first and second operation with “choice of an element from set A ” and “choice of an element from set B ,” we obtain the following consequence of Theorem 3.2.1:

Theorem 3.2.2 (Multiplication Rule) *If A_1 and A_2 are finite sets consisting, respectively, of k_1 and k_2 elements, then the Cartesian product $A_1 \times A_2$ consists of $k_1 k_2$ elements.*

Theorem 3.2.2 allows for an immediate generalization, we can define Cartesian products of more than two sets. Thus, if $A_1, \dots, A_n$ are some sets, then their Cartesian product $A_1 \times A_2 \times \dots \times A_n$ is the set of all n -tuples $(a_1, a_2, \dots, a_n)$ with $a_i \in A_i, i = 1, \dots, n$. By easy induction, Theorem 3.2.2 can now be generalized as follows:

Theorem 3.2.3 *If $A_1, \dots, A_n$ are finite, with A_i consisting of k_i elements ($i = 1, \dots, n$), then $A_1 \times \dots \times A_n$ contains $k_1 \cdots k_n$ elements.*

■ EXAMPLE 3.2

The total number of possible initials consisting of three letters (name, middle name, family name) is 26^3 . Each three-letter initial is an element of the set $A \times A \times A$, where A is the alphabet, so $k_1 = k_2 = k_3 = 26$. The total number of possible two- or three-letter initials is the number of the elements in the union $(A \times A) \cup (A \times A \times A)$, equal to $26^2 + 26^3 = 18,252$.

■ EXAMPLE 3.3 License Plates

Most states now use the system where a license plate has six symbols. One type (call it A) of such licenses has a prefix of three letters followed by a three-digit number (e.g., CQX 786). Other states use system (call it B) with a two-letter prefix, followed by a four-digit number (e.g., KN 7207). Still other states use system C, a digit and two letters, followed by a three digit number (e.g., 2CP 412). In addition states allow (for a special fee) “personalized” plates such as CATHY3 or MYCAR. Disregarding the personalized plates, which type of the license plate system can register most cars?

SOLUTION. Let A and D stand for the alphabet and for the set of 10 digits: $0, 1, \dots, 9$. Then, a license plate from system A can be regarded as an element of $A \times A \times A \times D \times D \times D$, while a license plate in system B is an element of the set $A \times A \times D \times D \times D \times D$. The numbers of elements in these Cartesian products are $26^3 \times 10^3$ and $26^2 \times 10^4$. The ratio is $26/10 = 2.6$, so in the state using an A system 2.6 times more cars can be registered than in the state with a B system.

Regarding system C, the answer depends whether or not 0 is allowed in the prefix. If the plate such as 0HY 314 is not allowed (e.g., because the digit 0 can be confused with the letter O), then the number of possible license plates is only $9 \times 26 \times 26 \times 10 \times 10 \times 10$, which is 10% less than the number of plates possible in states using system B. If 0 is allowed as the first character, then the numbers of plates of types B and C are the same.

In Examples 3.1 through 3.3 the set of ways of performing the second operations is the same regardless of which option was selected for the first operation. However, Theorem 3.2.1 remains true if the sets of ways of performing the second operation depend on the choice of the first operation. In particular, we can think of the first and second operation as two consecutive choices of an element from *the same set*, without returning the chosen elements. If the set, say A , has n elements, then the first operation (choice of an element) can be performed in n ways. If the chosen element is not returned, then the second choice can be performed in $n - 1$ ways only, and we have the following:

Corollary 3.2.4 *The number of ordered pairs (x, y) with $x \neq y$ that can be formed out of n distinct elements of a set of size n is $n(n - 1)$.*

Instead of thinking in terms of operations, we can still use Cartesian products here. Thus, $A \times A$ has n^2 elements by Theorem 3.2.3, of which n are of the form (x, x) . The number of pairs with elements distinct is $n^2 - n = n(n - 1)$.

We can generalize these considerations as follows:

Definition 3.2.1 An ordered sequence of k elements selected without replacement from a set of n distinct elements ($n \geq k$) is called a permutation of k out of n elements. $\square$

Theorem 3.2.5 *The number of permutations of k out of n , denoted P_n^k , equals*

$$P_n^k = n(n - 1) \cdots (n - k + 1).$$

Proof: The argument here repeatedly uses the “operation” principle: the first choice can be made in n ways, the second in $n - 1$ ways, the k th in $n - (k - 1) = n - k + 1$ ways. $\square$

If $k = n$, consecutive choices form an ordering of the entire set of size n . We obtain the following:

Corollary 3.2.6 *The set of n elements can be ordered in*

$$P_n^n = n(n-1) \times \cdots \times 2 \times 1 \quad (3.2)$$

distinct ways.

The product (3.2) occurs often and has a special symbol:

$$n! = 1 \times 2 \times \cdots \times (n-1)n$$

to be read “ n factorial.” We have therefore

$$P_n^k = \frac{n!}{(n-k)!}. \quad (3.3)$$

For a reason that will become apparent later, we adopt the convention

$$0! = 1. \quad (3.4)$$

■ EXAMPLE 3.4

The letters I, I, I, I, M, P, P, S, S, S, S are arranged at random. What is the probability that the arrangement will spell MISSISSIPPI?

SOLUTION. We can solve this problem treating the choices of consecutive letters as “operations.” The first operation must give the letter M; hence, there is only one way of choosing it. The next letter (out of the remaining 10) must be an I, and it can be selected in four ways. Proceeding in this way, the sequence of consecutive 11 choices leading to the word MISSISSIPPI can be performed in $1 \times 4 \times 4 \times 3 \times 3 \times 2 \times 1 \times 2 \times 2 \times 1 \times 1$ ways, which equals $4!4!2!1!$. On the other hand, the total number of ways one can perform the operations of consecutively choosing letters from the set is $11!$. Consequently, the required probability equals

$$p = \frac{4!4!2!1!}{11!}. \quad (3.5)$$

In this solution, the letters are regarded as distinguishable, as if we had four letters S , labeled S_1, S_2, S_3 , and S_4 , and similarly for the other letters. In this case, the numerator and denominator are, respectively, the number of ways one can order the set of distinguishable letters so as to form the word MISSISSIPPI and the total number of orderings. Alternatively, one can regard the identical letters as indistinguishable, and in this case, we have only one way of ordering them so as to spell the required word, and a total of $11!/(4!4!2!1!)$ distinguishable ways of ordering these letters. Indeed, the denominator here represents the number of ways of permuting letters so as to leave the arrangement invariant. Now,

$$p = \left(\frac{11!}{4!4!2!1!} \right)^{-1},$$

which is the same as (3.5).

■ EXAMPLE 3.5 Birthday Problem

The following problem has a long tradition and appears in many textbooks. If r randomly chosen persons attend a party, what is the probability p_r that none of them will have a birthday on the same day?

SOLUTION. Here we make the following assumption: (1) all years have 365 days (i.e., leap years are disregarded), (2) each day is equally likely to be a birthday of a person (i.e., births occur uniformly throughout the year), and (3) no twins attend the party. To compute p_r , we must find the number of all possible ways in which birthdays can be allocated to r people, and the number of such allocations in which birthdays do not repeat. The first number is 365^r (by Theorem 3.2.3), while the second number is P_{365}^r (assuming $r \leq 365$; if $r > 365$, we must have at least one birthday repeating, so $p_r = 0$). Thus, for $r \leq 365$, we have

$$\begin{aligned} p_r &= \frac{P_{365}^r}{365^r} = \frac{365}{365} \times \frac{365-1}{365} \times \cdots \times \frac{365-r+1}{365} \\ &= \left(1 - \frac{1}{365}\right) \left(1 - \frac{2}{365}\right) \cdots \left(1 - \frac{r-1}{365}\right). \end{aligned}$$

As a first approximation, neglecting all products which have denominators of order 365^2 or higher, we can take

$$p_r \approx 1 - \frac{1+2+\cdots+(r-1)}{365} = 1 - \frac{(r-1)r}{730}. \quad (3.6)$$

This approximation works quite well for small r . Now since $\log(1-x) \approx -x$, we have

$$\log p_r \approx \frac{-r(r-1)}{730} \quad \text{and consequently} \quad p_r \approx e^{-\frac{r(r-1)}{730}}. \quad (3.7)$$

It is interesting that for $r = 23$ a repeated birthday is about as likely as no repetition. The smallest r for which p_r is less than 0.01 is 56.

PROBLEMS

- 3.2.1** A certain set contains n distinct elements. Find n if the number of: **(i)** All possible permutations of length 2 equals 90. **(ii)** Permutations of length 3 is 10 times larger than the number of permutations of length 2.
- 3.2.2** A skyscraper is 40 stories tall. Five people enter the elevator on the first floor. Assuming each person is equally likely to get off at any of the 39 floors 2, 3, ..., 40, what is the probability that all people will get off at different floors? Find the exact value, and then derive and compute the approximations analogous to (3.6) and (3.7).
- 3.2.3** A two letter code is to be formed by selecting (without replacement) the letters from a given word. Find the number of possible codes if the word is: **(i)** CHART. **(ii)** ALOHA. **(iii)** STREET.
- 3.2.4** Determine the number of 0s at the end of $16!$ and $27!$.
- 3.2.5** Seated at random in a row of n seats are n people, among them John and Mary. Find the probability that: **(i)** John sits next to Mary. **(ii)** John sits next to Mary on her right. **(iii)** John sits somewhere to the right of Mary. **(iv)** John and Mary sit exactly two seats apart.
- 3.2.6** Seated at random at a round table with n seats are n people, among them John and Mary. **(i)** Answer questions (i)–(iv) of Problem 3.2.5. Anything peculiar about the answer to (iii)? **(ii)** Assume that $n = 2k$. Find the probability that John and Mary sit facing each other (e.g., numbers 1 and 7 on the clock). **(iii)** Assume that $n = 8$. Find the probability that good friends, Nico, Noah, and Helen, are not separated by other guests.

- 3.2.7** Five men and five women are to be seated in a row of 10 chairs. Find the number of possible arrangements if: **(i)** Men are required to sit in alternating seats. **(ii)** No two men are to be seated next to each other.
- 3.2.8** A total of 12 girls and 17 boys go to a dance. **(i)** How many possible dancing pairs (boy–girl) may be formed? **(ii)** The dance floor can accommodate at most 11 pairs at a time. If each dance lasts 10 minutes and is followed by a 2-minute break, how much time, at least, will elapse before each boy will have danced with each girl at least once? **(iii)** Answer the same question as in (ii) if the dance floor can accommodate 15 pairs at a time.
- 3.2.9** Susan has five dresses, three skirts, four blouses, three pairs of shoes, and two hats. She always wears shoes and either a dress or a blouse and a skirt. She may or may not wear a hat. **(i)** How many different combinations can she wear? **(ii)** Suppose Susan can afford buying either a dress or a hat (but not both). What should she buy to maximize the number of different combinations that she can wear? **(iii)** Suppose that Susan's brown shoes do not match her pink or blue dress and that the blue hat does not match her yellow blouse. How many matching combinations can she wear?
- 3.2.10** A restaurant menu has 5 appetizers, 3 soups, 15 entrees, and 3 desserts. **(i)** Assuming you are going to order one item from each group, how many possible dinners can you order? **(ii)** Assume you are at the restaurant with a friend. How many different orders for two full dinners can you place if your friend's choice of every item is not necessarily the same as yours? **(iii)** Answer the question in part (ii) under the constraint that you do not order the same entree and the same dessert as your friend (but the soup and/or appetizer may be the same).
- 3.2.11** Find the percentage of six-digit numbers that have all digits distinct.
- 3.2.12** A regular die is tossed n times. Find the probability that: **(i)** Each side turns up exactly once if $n = 6$. **(ii)** Each side turns up at least once if $n = 8$.
- 3.2.13** Find the number of three-digit integers (i.e., integers between 100 and 999) that have all digits distinct. How many of them are odd?
- 3.2.14** Let e_r be the probability that exactly two people in a group of r have the same birthday, and let p_r be the probability that everybody in the group has a different birthday. Show that

$$e_r = \frac{r(r-1)}{2} p_{r-1}.$$

3.3 BINOMIAL COEFFICIENTS

The permutations discussed in Section 3.2 were ordered selections from a certain set. Often, these orders are irrelevant, as we are interested only in the total number of possible choices. Such choices are referred to as *combinations*.

Definition 3.3.1 A subset of size k selected from a set of size n (regardless of the order in which this subset was selected) is called a *combination* of k out of n . $\square$

Theorem 3.3.1 The number of combinations of k out of n , C_n^k , is given by

$$C_n^k = \frac{P_n^k}{k!}. \quad (3.8)$$

Proof: By Theorem 3.2.5 we have P_n^k different permutations of k out of n elements. Each permutation determines the set of k elements selected and their order. Consequently, $k!$ permutations lead to the same combination, which proves (3.8). $\square$

The ratio $P_n^k/k!$ appears in various contexts, and it is convenient to have a special symbol for it.

Definition 3.3.2 The ratio

$$\frac{P_n^k}{k!} = \frac{n(n-1) \cdots (n-k+1)}{k!} \quad (3.9)$$

is called a *binomial coefficient* and is denoted by $\binom{n}{k}$, to be read as “ n choose k .” $\square$

Using (3.3), we have

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \quad (3.10)$$

Observe, however, that (3.10) requires n to be an integer, whereas in (3.9) n can be any real number (k has to be an integer in both cases).

■ EXAMPLE 3.6

As an illustration, let us evaluate $\binom{-1/2}{k}$, which we will use later. We have

$$\begin{aligned} \binom{-1/2}{k} &= \frac{(-\frac{1}{2})(-\frac{1}{2}-1)(-\frac{1}{2}-2) \cdots (-\frac{1}{2}-k+1)}{k!} \\ &= \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2}) \cdots (-\frac{2k-1}{2})}{k!} = \frac{(-1)^k 1 \times 3 \times 5 \times (2k-1)}{2^k k!}. \end{aligned}$$

Multiplying the numerator and denominator by $2 \times 4 \times \cdots \times (2k) = 2^k k!$, we get

$$\binom{-1/2}{k} = \frac{(-1)^k}{2^{2k}} \binom{2k}{k}. \quad (3.11)$$

In this section, we tacitly assume that n is an integer with $n \geq k$.

Observe also that the symbol $\binom{n}{k}$ makes sense for $k = 0$ and $k = n$, in view of the convention that $0! = 1$. Thus, we have

$$C_n^k = \binom{n}{k} \quad (3.12)$$

for all integers k and n such that $n \geq k$. For $k = 0$, we have $C_n^0 = 1$, since there is only one empty set, and $C_n^n = 1$, since only one set of size n can be selected out of a set of size n . Formula (3.12) gives correct values, namely

$$\binom{n}{n} = \binom{n}{0} = 1. \quad (3.13)$$

We will now study some properties of the binomial coefficients $\binom{n}{k}$. First,

$$\binom{n}{k} = \binom{n}{n-k}, \quad (3.14)$$

which follows from the symmetry in formula (3.10). One can also prove (3.14) by observing that choosing a set of size k is equivalent to “leaving out” a set of size $n - k$. The number of different sets of size k chosen must be equal to the number of different sets of size $n - k$ “chosen” by leaving them out.

We will now prove the following theorem:

Theorem 3.3.2 (Pascal’s Triangle) *The binomial coefficients satisfy the relation*

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}. \quad (3.15)$$

Proof: The formula can be easily proved by expressing the left-hand side using (3.8) and reducing it algebraically to get the right-hand side. It is, however, more instructive to rely on the interpretation of the coefficients $\binom{n}{k}$ as C_n^k . The right-hand side of (3.15) counts the number of sets of size k that can be chosen out of a set of size $n + 1$. Let us take one element of the latter set and label it somehow. We have then a set of n unlabeled and 1 labeled element. Each subset of size k is one of the following two categories: (1) subsets that contain only k unlabeled elements, or (2) subsets that contain $k - 1$ unlabeled elements and one labeled element. The two terms on the left-hand side of (3.15) count the numbers of subsets of the first category and of the second category. $\square$

The name *Pascal’s triangle* reflects a way of obtaining the coefficients $\binom{n}{k}$. We build Pascal’s triangle starting with the top row (counted as the zeroth row), which consists of the single number 1 (see Figure 3.1). Then we obtain each number in the subsequent rows as a sum of two numbers directly above it (as marked with arrows in the fifth row). The consecutive numbers in the n th row are, reading from the left, the values of

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots$$

so that, for example, $\binom{6}{3} = 20$, as marked on the triangle in Figure 3.1. While the Pascal triangle was very useful in the past, today it is of a historical value as statistical packages are used to obtain values of binomial coefficients.

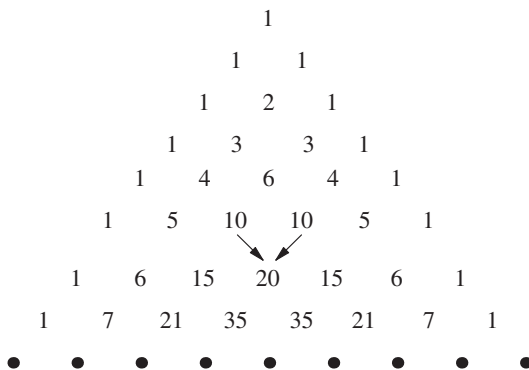


Figure 3.1 Pascal’s triangle.

The name *binomial coefficient* is connected to the following formula:

Theorem 3.3.3 (Newton's Binomial Formula) *For any positive integer n and real x, y , we have*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k. \quad (3.16)$$

Proof: We will prove the theorem by induction. For $n = 1$, the right-hand side equals $\binom{1}{0}x + \binom{1}{1}y = x + y$. Assume now that the assertion holds for some n , and multiply both sides of (3.16) by $(x + y)$. Then

$$\begin{aligned} (x + y)^{n+1} &= (x + y) \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \\ &= \sum_{k=0}^n \binom{n}{k} x^{n+1-k} y^k + \sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k+1} \\ &= \sum_{k=0}^n \binom{n}{k} x^{n+1-k} y^k + \sum_{k=1}^{n+1} \binom{n}{k-1} x^{n-(k-1)} y^k. \end{aligned}$$

Separating the term for $k = 0$ in the first sum, and the term for $k = n + 1$ in the last sum, we may write

$$\begin{aligned} (x + y)^{n+1} &= x^{n+1} + \sum_{k=1}^n \left[\binom{n}{k} + \binom{n}{k-1} \right] x^{(n+1)-k} y^k + y^{n+1} \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k} x^{(n+1)-k} y^k, \end{aligned}$$

where the last equality is due to Theorem 3.3.2. □

The following theorem is an immediate consequence of Theorem 3.3.3 applied to $(1 + 1)^n$ and $(1 - 1)^n$, respectively.

Theorem 3.3.4 *The binomial coefficients satisfy the identities*

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n \quad (3.17)$$

and

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots \pm \binom{n}{n} = 0. \quad (3.18)$$

We also have the following theorem:

Theorem 3.3.5 *For every $n = 1, 2, \dots$ and every $k = 0, 1, \dots, n$*

$$\binom{n}{0} \binom{n}{k} + \binom{n}{1} \binom{n}{k-1} + \cdots + \binom{n}{k} \binom{n}{0} = \binom{2n}{k}. \quad (3.19)$$

Proof: Consider the product $(1+x)^n(1+x)^n = (1+x)^{2n}$. Expanding the right-hand side, we obtain

$$\sum_{k=0}^{2n} \binom{2n}{k} x^k, \quad (3.20)$$

while the left-hand side equals

$$\left[\sum_{i=0}^n \binom{n}{i} x^i \right] \times \left[\sum_{j=0}^n \binom{n}{j} x^j \right]. \quad (3.21)$$

For $k \leq n$, comparison of the coefficients of x^k in (3.20) and (3.21) gives (3.19). $\square$

As a consequence of (3.19), we obtain a corollary:

Corollary 3.3.6

$$\sum_{j=0}^n \binom{n}{j}^2 = \binom{2n}{n}.$$

Proof: Take $k = n$ in (3.19) and use the fact that

$$\binom{n}{n-i} = \binom{n}{i}.$$

$\square$

Below we present some examples of the use of binomial coefficient in solving various probability problems, some with a long history.

■ **EXAMPLE 3.7**

Let us consider a selection without replacement from a finite set containing two categories of objects. If n balls are to be selected from an urn containing r red and b blue balls, one might want to know the probability that there will be exactly k red balls chosen.

SOLUTION. We apply here the “classical” definition of probability. The choice of n objects without replacement is the same as choosing a subset of n objects from the set of total of $r+b$ objects. This can be done in $\binom{r+b}{n}$ different ways. Since we must have k red balls, this choice can be made in $\binom{r}{k}$ ways. Similarly, $n-k$ blue balls can be selected in $\binom{b}{n-k}$ ways. As each choice of k red balls can be combined with each of the $\binom{b}{n-k}$ choices of blue balls then, by Theorem 3.2.2, the total number of choices is the product $\binom{r}{k} \binom{b}{n-k}$ and

$$P(\text{exactly } k \text{ red balls}) = \frac{\binom{r}{k} \binom{b}{n-k}}{\binom{r+b}{n}}. \quad (3.22)$$

The next example shows an interesting application of formula (3.22).

■ **EXAMPLE 3.8**

Consider the problem of estimating the number of fish in a lake (the method described below is also used to estimate the sizes of bird or wildlife populations). The lake contains an unknown number N of fish. To estimate N , we first catch c fish, label them, and release them back into the lake. We assume here that labeling does not harm fish in any way, that the labeled fish mix with unlabeled ones in a random manner, and that N remains constant (in practice, these assumptions may be debatable). We now catch k fish, and observe the number, say x , of labeled ones among them. The values c and k are, at least partially, under the control of the experimenter. The unknown parameter is N , while x is the value occurring at random, and providing the key to estimating N . Let us compute the probability $P_N = P_N(x)$ of observing x labeled fish in the second catch if there are N fish in the lake. We may interpret fish as balls in an urn, with labeled and unlabeled fish taking on the roles of red and blue balls. Formula (3.22) gives

$$P_N = \frac{\binom{c}{x} \binom{N-c}{k-x}}{\binom{N}{k}}. \quad (3.23)$$

To estimate N , we can use the principle of *maximum likelihood* (to be explored in detail in Chapter 11). At present, it suffices to say that this principle suggests using as $\hat{N}$, an estimator of N , the value of N that maximizes (3.23). Let us call this value $\hat{N}$. Let us call this value $\hat{N}$. It depends on the observed value x and hence is itself random. Thus, $\hat{N}$ is defined by the condition

$$P_{\hat{N}} \geq P_N \quad \text{for all } N$$

and our objective is to find the maximizer of P_N . Since N is a discrete variable, we cannot use methods of finding a maximum based on derivatives. Instead, the method that works in this case is based on the observation that if the function P_N has a maximum (possibly local) at N^* , then $P_{N^*}/P_{N^*-1} > 1$ and $P_{N^*+1}/P_{N^*} < 1$. If two neighboring probabilities have equal values, the ratio equals 1. Consequently, we should study the ratio P_N/P_{N-1} and find all arguments at which this ratio crosses the threshold 1. After some reduction, we have

$$P_N/P_{N-1} = \frac{(N-c)(N-k)}{N(N-c-k+x)}.$$

The above ratio always exceeds 1 if $x = 0$, so in this case the maximum is not attained. Assume now that $x > 0$. The inequality

$$P_N/P_{N-1} \geq 1$$

is equivalent to

$$N \leq \frac{kc}{x}, \quad (3.24)$$

with the equality occurring if and only if $P_N/P_{N-1} = 1$. Thus, the maximum is attained at⁶

$$\hat{N} = \left\lceil \frac{kc}{x} \right\rceil,$$

and also at $kc/x - 1$ if the latter value is an integer. Let us observe that the result above is consistent with common intuition: The proportion of labeled fish in the whole lake

⁶The integer part of a , $[a]$, is the largest integer not exceeding a ($[3.21] = 3$, $[-1.71] = -2$, etc.).

is c/N , and it should be close to the proportion x/k of labeled fish in the second catch. This gives the approximate equation $c/N \approx x/k$, with the solution $N \approx kc/x$.

■ EXAMPLE 3.9

To supplement their revenues, many states are sponsoring number games or lotteries. The details vary from state to state, but generally, a player who buys a lottery ticket chooses several numbers from a specified set of numbers. We will carry the calculations for the choice of 6 out of 50 numbers 1, 2, ..., 50, which is quite typical. After the sales of tickets close, six winning numbers are chosen at random from the set 1, 2, ..., 50. All those (if any) who chose six winning numbers share the Big Prize; if there are no such winners, the Big Prize is added to the next week's Big Prize. Those who have five winning numbers share a smaller prize, and so on. Let $P(x)$ be the probability that a player has exactly x winning numbers. We will compute $P(x)$ for $x = 6, 5, 4$, and 3. The calculations would be the same if the winning numbers were chosen in advance, but remained secret to the players. We can now represent the situation in a familiar scheme of an urn with 6 winning numbers and 44 losing numbers, and the choice of 6 numbers from the urn (without replacement). This is the same problem as that of labeled fish. The total number of choices that can be made is $\binom{50}{6}$, while $\binom{6}{x} \binom{44}{6-x}$ is the number of choices with exactly x winning numbers. Thus,

$$P(x) = \frac{\binom{6}{x} \binom{44}{6-x}}{\binom{50}{6}}.$$

For $x = 6$, we have

$$P(6) = \frac{1}{\binom{50}{6}} = \frac{6!}{50 \times 49 \times 48 \times 47 \times 46 \times 45} = \frac{1}{15,890,700} = 6.29 \times 10^{-8}.$$

Similarly $P(5) = 1.66 \times 10^{-5}$, $P(4) = 8.93 \times 10^{-4}$ and $P(3) = 0.016669$.

Thus, the chances of winning a share in the Big Prize are about 1 in 16 million. It would therefore appear that there should be, on average, one big winner in every 16 million tickets sold. The weekly numbers of tickets sold are well known, and it turns out that the weekly numbers of winners (of the Big Prize) vary much more than one would expect. For example, in weeks where the number of tickets sold is about 16 million, one could expect no winner, one winner, or two winners; three winners is unlikely. In reality, it is not at all uncommon to have five or more winning tickets in a week with 16 million tickets sold. These observations made some people suspicious about the honesty of the process of drawing the numbers, to the extent that there have been attempts to bring suit against the lottery (e.g., accusing the organizers of biasing the lottery balls with certain numbers so as to decrease their chance of being selected, thus favoring some other numbers).

Actually, the big variability of weekly numbers of winners is to be expected if one realizes that these numbers depend on *two* chance processes: the choice of winning numbers from the urn (which may be, and probably is, quite fair) and the choice of numbers by the players. This choice is definitely not uniform. It favors certain combinations, which seem more "random" to the naive persons than other choices. For instance, the combination 1, 2, 3, 4, 5, 6 appears less likely than (say) 5, 7, 19, 20, 31, and 45. As a consequence some combinations are selected more often by the players than others. Each combination has the same chance of being the winning one, but some may have higher numbers of winners associated with them. This point can be illustrated by

the following analogy: Imagine that each week a first name (Mary, Susan, George, etc.) is chosen at random from the set of all names used, and all persons in the state with the selected name share the prize. The chances of being chosen are the same for John as for Sebastian, as they depend on the process of sampling names of winners. But if the name Sebastian is chosen, each Sebastian will share the prize with many fewer other winners than if the name John were selected. Here the numbers of winners to share the prize depend on another process, namely that of parents selecting names for their children.

■ EXAMPLE 3.10

We have k urns, labeled $1, \dots, k$, and n identical (indistinguishable) balls. In how many ways can these balls be distributed in k urns?

SOLUTION. There are no restrictions here on the number of balls in an urn, or the number of empty urns. To get the answer, let us identify each possible allocation with a string of $k + 1$ bars and n circles, of the form

$$| \circ \circ || \circ \circ \circ \circ | \circ | \cdots \circ |,$$

with the only condition being that the string should start and end with a bar. The spaces between bars represent urns. Thus, in the arrangement above, the first urn contains 2 balls, the second none, the third 4 balls, and so on. Clearly, the number of distinct arrangements equals $\binom{n+k-1}{n}$ —the number of distinct arrangements of $k - 1$ bars and n circles. Indeed, we have a string of $n + k - 1$ symbols (not counting the two extreme bars), and each arrangement is obtained by specifying n places for the symbol $\circ$.

Example 3.10 shows that the binomial coefficient can be interpreted in two ways. On the one hand, $\binom{a+b}{a}$ is the number C_{a+b}^a of distinct sets of size a that can be chosen out of a set of size $a + b$. On the other hand, $\binom{a+b}{a}$ is also the number of distinct strings of a indistinguishable elements of one kind and b indistinguishable elements of another kind. To see this, it suffices to think of the string as being determined by the choice of “ a out of total of $a + b$ ” slots into which we assign elements of the first kind.

■ EXAMPLE 3.11 Matching Problem

A secretary typed n letters and addressed n envelopes. For some reason, the letters were put into envelopes at random. What is the probability of at least one match, that is, of at least one letter being put into the correct envelope?

SOLUTION. This problem appears in many textbooks under various formulations (e.g., of guests receiving their hats at random). One could expect the probability of at least one match to vary greatly with n . However, the contrary is true: this probability is almost independent of n . Let A_i be the event that i th letter is placed in the correct envelope. Using formula (2.6), we have

$$\begin{aligned} P(\text{at least one } A_i) &= P(A_1 \cup \cdots \cup A_n) \\ &= \sum P(A_i) - \sum_{i < j} P(A_i \cap A_j) \\ &\quad + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots \pm P(A_1 \cap \cdots \cap A_n). \end{aligned}$$

By symmetry, the probability of each intersection depends only on the number of events in the intersection,⁷ so we let p_r denote the probability of the intersection of r events, $p_r = P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_r})$. Clearly, the numbers of terms in the consecutive sums are

$$\binom{n}{1}, \binom{n}{2}, \binom{n}{3}, \dots$$

and

$$P(\text{at least one } A_i) = \binom{n}{1} p_1 - \binom{n}{2} p_2 + \binom{n}{3} p_3 - \cdots \pm \binom{n}{n} p_n. \quad (3.25)$$

To evaluate p_r , we can argue as follows: Assume that the envelopes are ordered in some way. The total number of ways one can order n letters is $n!$. If specific r events, say $A_{i_1}, \dots, A_{i_r}$, are to occur (perhaps in conjunction with other events), then the letters number $i_1, \dots, i_r$ must be at their appropriate places in the ordering (to match their envelopes). The remaining $n - r$ letters can appear in any of the $(n - r)!$ orders. Thus,

$$p_r = \frac{(n - r)!}{n!}.$$

Consequently, the r th term in the sum (3.25) equals (up to the sign)

$$\binom{n}{r} \frac{(n - r)!}{r!} = \frac{1}{r!},$$

and we obtain

$$P(\text{at least one match}) = \frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \cdots \pm \frac{1}{n!}.$$

Since

$$e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!},$$

we have

$$P(\text{at least one match}) \approx 1 - \frac{1}{e},$$

with the accuracy increasing as $n \rightarrow \infty$. The approximation is actually quite good for small n . The limiting value is 0.63212056, while the exact values of the probability π_n of at least one match for selected values of n are

$$\pi_1 = 1$$

$$\pi_2 = 1 - \frac{1}{2} = 0.5$$

$$\pi_3 = 1 - \frac{1}{2} + \frac{1}{6} = 0.6666667$$

$$\pi_4 = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} = 0.625$$

$$\pi_5 = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{120} = 0.6333333$$

$$\pi_6 = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{120} - \frac{1}{720} = 0.6319444$$

⁷This property, called exchangeability of events, will be discussed in more detail in Section 4.6.

$$\pi_7 = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{120} - \frac{1}{720} + \frac{1}{5040} = 0.6321429$$

$$\pi_8 = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{120} - \frac{1}{720} + \frac{1}{5,040} - \frac{1}{40,320} = 0.6321181.$$

■ EXAMPLE 3.12 Ballot Problem

Suppose that in an election, candidate A receives a votes, while candidate B receives b votes, where $a > b$. Assuming that votes are counted in random order, what is the probability that during the whole process of counting A will be ahead of B?

SOLUTION. Note that other votes, if any, do not matter, and we may assume that $a + b$ is the total number of votes. The process of counting votes is determined by the arrangement of the votes, that is, the arrangement of a symbols A, and b symbols B. Clearly, such an arrangement is uniquely determined by specifying the locations of the As (or, equivalently, Bs). It might be helpful to use a graphical representation: define the function $C(k)$ as the net count for candidate A after inspection of k votes. Thus, if in the first k votes, we had r votes for A and $k - r$ votes for B, then $C(k) = r - (k - r) = 2r - k$. We can then represent the process of counting as a polygonal line that starts at the origin and has vertices $(k, C(k))$, $k = 1, \dots, a + b$ (see Figure 3.2).

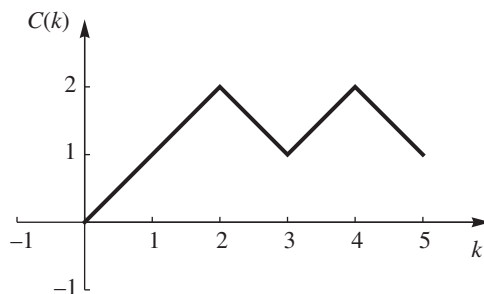


Figure 3.2 Process of counting votes.

In Figure 3.2, we have the beginning of counting, when the first five votes inspected are AABAB. The problem can now be formulated as finding the probability that the counting function $C(x)$ lies above the x -axis for all $x = 1, 2, \dots, a + b$. Observe that the first vote counted must be for A (as in Figure 3.2); this occurs with probability $a/(a + b)$.

The remaining votes will give a polygonal line leading from $(1, 1)$ to $(a + b, a - b)$, and we must find the number of such lines that will never touch or cross the x -axis. The number of such lines is equal to the total number of lines from $(1, 1)$ to $(a + b, a - b)$ minus the number of lines from $(1, 1)$ to $(a + b, a - b)$ which touch or cross the x -axis. The total number of lines leading from $(1, 1)$ to $(a + b, a - b)$ is $\binom{a+b-1}{a-1}$, since each such line has $a - 1$ steps “up” and b steps “down,” which can be ordered in any manner. Thus, it remains to count the number of lines from $(1, 1)$ to $(a + b, a - b)$ that touch or cross the x -axis. Let V be the set of all such lines. Each line in V must touch the x -axis for the first time at some point, say t (see Figure 3.3). If we reflect the part of

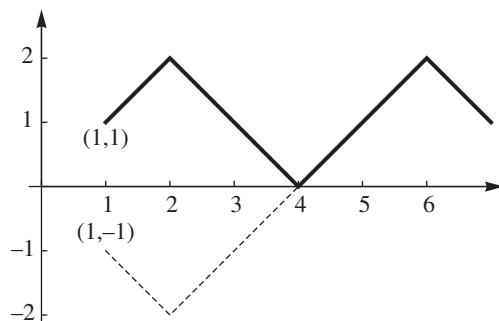


Figure 3.3 Reflection principle.

this line that lies to the left of t with respect to x -axis, we obtain a line leading from $(1, -1)$ to $(a + b, a - b)$. Moreover, different lines in V will correspond to different lines leading from $(1, -1)$ to $(a + b, a - b)$, and each line in the latter set will be obtained from some line in V . This means that the set V has the same number of lines as the set of lines leading from $(1, -1)$ to $(a + b, a - b)$. But the latter set contains $\binom{a+b-1}{a}$ lines, since each such line must have a steps “up” and $b - 1$ steps “down.” Consequently, the required probability equals

$$p = \frac{a}{a+b} \times \frac{\binom{a+b-1}{a-1} - \binom{a+b-1}{a}}{\binom{a+b-1}{a-1}} = \frac{a-b}{a+b}.$$

■ EXAMPLE 3.13 Poker

We now consider the probabilities of several poker hands (some students will probably say that finally the book gives some useful information).

In poker, five cards are dealt to a player from a standard deck of 52 cards. The number of possible hands is therefore $\binom{52}{5} = 2,598,960$. The lowest type of hand is that containing one pair (two cards of the same denomination, plus three unmatched cards). To find the number of possible hands containing one pair, one can think in terms of consecutive choices leading to such a hand:

- The denomination of the cards in a pair can be chosen in $\binom{13}{1}$ ways.
- The suits of the pair can be chosen in $\binom{4}{2}$ ways.
- The choice of denominations of the remaining three cards can be made in $\binom{12}{3}$ ways.
- The suits of those three cards may be chosen in 4^3 ways. Altogether, combining (a)–(d), we have

$$P(\text{one pair}) = \frac{\binom{13}{1} \binom{4}{2} \binom{12}{3} 4^3}{\binom{52}{5}}.$$

The next kind of hand is the one containing two pairs. Here the argument is as follows:

- The denominations of the two pairs can be selected in $\binom{13}{2}$ ways.
- The suits of cards in these two pairs can be selected in $\binom{4}{2} \times \binom{4}{2}$ ways.
- The remaining card may be chosen in $\binom{11}{1} \times 4$ ways (two denominations are eliminated).

Combining (a)–(c), we have

$$P(\text{two pairs}) = \frac{\binom{13}{2} \binom{4}{2}^2 \binom{11}{1} \times 4}{\binom{52}{5}}.$$

Finally, we calculate the probability of a straight (probabilities of remaining hands are left as exercise). A straight is defined as a hand containing five cards in consecutive denominations but not of the same suit (e.g., 9, 10, jack, queen, and king). An ace can appear at either end, so we could have a straight of the form ace, 2, 3, 4, 5, as well as 10, jack, queen, king, ace.

The number of hands with a straight can be computed as follows: Each such hand is uniquely determined by the lowest denomination (ace, 2, 3, ..., 10) in 10 ways. Then, the suits of five cards are chosen in $4^5 - 4$ ways: 4^5 is the total number of choices of suits, and we subtract 4 selections in which all cards are of the same suit. Thus,

$$P(\text{straight}) = \frac{10(4^5 - 4)}{\binom{52}{5}}.$$

PROBLEMS

- 3.3.1 (i)** A committee of size 50 is to be formed out of the US Senate at random. Find the probability that each state will be represented. **(ii)** If a committee of size k is to be formed out of the US Senate find how large must k be in order for the event “at least one senator from Ohio is included” to be more likely than the event “no senator from Ohio is included.”
- 3.3.2** A shipment of 30 items is received. For the quality control, three items are randomly selected, and if more than one of them is defective then the whole shipment is rejected. Find the probability that the shipment will be accepted if it has: **(i)** Three defective items **(ii)** Ten defective items.
- 3.3.3** How many ways can one order the deck of 52 cards so that all four kings are next to each other?
- 3.3.4** Peter lives at the corner of 2nd Avenue and 72nd Street. His office is in the building at a corner of 7th Avenue and 78th Street. The streets and avenues in the city form a perpendicular grid, with no streets or passages in the middle of the blocks. Peter walks to work along either street or avenue, always in the direction that gets him closer to his office. He always returns home by subway, so he walks across town only once a day. **(i)** How many different paths can Peter choose to go to work? **(ii)** If Peter makes a list of all possible paths and chooses one of them randomly every morning, how likely it is that he will not walk 4th Avenue between 75th and 76th streets during the next five working days?
- 3.3.5 (Poker Hands)** Find the probability of each of the following hands: **(i)** Royal flush (ace, king, queen, jack, and 10 in one suit), **(ii)** Straight flush (five cards of one suit in a sequence, but not a royal flush), **(iii)** Flush (five cards in one suit, but not a straight flush nor a royal flush), **(iv)** Four-of-a-kind (four cards of the same denomination), **(v)** Full house (one pair and one triple of the same denomination), **(vi)** Three-of-a-kind (three cards of the same denomination plus two cards unmatched).
- 3.3.6** Find the probability that a poker hand will contain two pairs (one red and the other black) and one unmatched card.

- 3.3.7** A poker player has $3\heartsuit, 7\diamond, 8\spadesuit, 9\heartsuit, Q\clubsuit$. He discards $3\heartsuit$ and $Q\clubsuit$ and obtains 2 cards.⁸
(i) What is the probability that he will have a straight? **(ii)** Answer the same question if $Q\clubsuit$ is replaced by $J\clubsuit$ (i.e., he discards $3\heartsuit$ and $J\clubsuit$).
- 3.3.8** A poker player has $3\heartsuit, 7\diamond, 8\heartsuit, 9\heartsuit, Q\clubsuit$. She discards $3\heartsuit$ and $Q\clubsuit$ and obtains 2 cards. What is the probability that she will have: **(i)** A straight flush. **(ii)** A flush, but not a straight flush. **(iii)** A straight, but not a straight flush.
- 3.3.9** A poker player has three-of-a-kind. He discards the two unmatched cards and obtains two new cards. Find the probability that he will have: **(i)** Three-of-a-kind. **(ii)** Four-of-a-kind. **(iii)** A full house.
- 3.3.10** **(i)** If n balls are put at random into n boxes, find the probability of exactly one box remaining empty? **(ii)** If n balls are randomly placed into k boxes ($n \geq k$), labeled $1, \dots, k$, find the probability that no box is empty.
- 3.3.11** Compute probabilities $P(x)$ of winning x numbers in lotteries, where the player chooses: **(i)** 5 out of 44 numbers. **(ii)** 6 out of 55 numbers.
- 3.3.12** Find the number of polygonal lines with vertices $(x, C(x))$, where $C(x)$ is as in Example 3.12 and with possible edges leading from $(x, C(x))$ to $(x+1, C(x)+1)$ or $(x+1, C(x)-1)$, connecting the points: **(i)** $(0, 0)$ and $(10, 0)$. **(ii)** $(0, 0)$ and $(10, 5)$. **(iii)** $(3, -2)$ and $(8, 1)$.
- 3.3.13** Find the number of polygonal lines (as in Problem 3.3.12) that join the points $(2, 3)$ and $(16, 5)$ and: **(i)** Never touch the x -axis. **(ii)** Never touch the line $y = 7$.

3.4 MULTINOMIAL COEFFICIENTS

Choosing a subset of size k out of a set of size n is logically equivalent to partitioning the set of size n into two subsets, one of size k and the other of size $n - k$. The number of such partitions is, by definition,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The theorem below generalizes this scheme.

Theorem 3.4.1 *Let $k_1, \dots, k_r$ be positive integers such that $k_1 + \dots + k_r = n$. The number of ways a set of n elements can be partitioned into r subsets of sizes $k_1, k_2, \dots, k_r$ equals*

$$\frac{n!}{k_1!k_2!\cdots k_r!}. \quad (3.26)$$

Proof: A partition above can be accomplished in steps: First, we choose k_1 out of n elements to form the first subset of the partition. Next, we choose k_2 elements out of the remaining $n - k_1$ elements, and so on, until we have $n - k_1 - k_2 - \dots - k_{r-2} = k_{r-1} + k_r$ elements, from which we choose k_{r-1} to form the next-to-last subset. The remaining k_r elements form the last subset. This can be accomplished, in view of Theorem 3.2.2, in

$$\binom{n}{k_1} \binom{n-k_1}{k_2} \binom{n-k_1-k_2}{k_3} \cdots \binom{n-k_1-\dots-k_{r-2}}{k_{r-1}} \quad (3.27)$$

ways. Simple algebra shows that formula (3.27) is the same as formula (3.26). $\square$

⁸The discarded cards are not mixed with the deck. Assume that the player receives the replacement of the discarded cards from the unused remainder of the deck.

The ratio (3.26) is called *multinomial coefficient* and is denoted by

$$\binom{n}{k_1, k_2, \dots, k_r}.$$

As a generalization of Newton's binomial formula, we have

Theorem 3.4.2 *For every integer n ,*

$$(x_1 + \dots + x_r)^n = \sum \binom{n}{k_1, k_2, \dots, k_r} x_1^{k_1} \dots x_r^{k_r}, \quad (3.28)$$

where the summation is extended over all subsets $(k_1, \dots, k_r)$ of nonnegative integers with $k_1 + \dots + k_r = n$.

Proof: In the product $(x_1 + \dots + x_r)^n$, one term is taken from each factor so that the general term of the sum has the form $x_1^{k_1} \dots x_r^{k_r}$ with $k_1 + \dots + k_r = n$. From Theorem 3.4.1, it follows that the number of times the product $x_1^{k_1} \dots x_r^{k_r}$ appears equals (3.26). $\square$

In an analogy with a formula (3.17), the sum of all multinomial coefficients equals r^n , which follows by substituting $x_1 = \dots = x_r = 1$ in (3.28).

The theorem is illustrated by the following example:

■ EXAMPLE 3.14

Suppose that one needs the value of the coefficient of $x^2 y^3 z^4 w$ in the expression $(x + y + z + w)^{10}$. One could argue that in the multiplication $(x + y + z + w) \times \dots \times (x + y + z + w)$ there are 10 factors, and each term will contain one component from each set of parentheses. Thus, choosing x from 2 out of 10 pairs of parentheses, y from 3 out of 10, and so on, amounts to partitioning 10 pairs of parentheses into four classes, with sizes $k_1 = 2, k_2 = 3, k_3 = 4$, and $k_4 = 1$. The total number of ways such a partition can be accomplished is the coefficient of $x^2 y^3 z^4 w$, and equals $\binom{10}{2, 3, 4, 1} = \frac{10!}{2!3!4!1!} = 12,600$.

An approximation to $n!$ is given by the so-called Stirling's formula.

Theorem 3.4.3 (Stirling's Formula) *We have*

$$n! \sim \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n}, \quad (3.29)$$

where the sign $\sim$ means that the ratio of the two sides tends to 1 as $n \rightarrow \infty$.

We shall not give the proof here, but interested readers can find it in more advanced texts, for example, in Chow and Teicher (1997).

■ EXAMPLE 3.15

A group of $2n$ boys and $2n$ girls is divided at random into two equal parts. What is the probability p_n and its approximation for large n , that boys and girls are divided evenly between the two groups?

SOLUTION. Clearly, the number of ways a group can be divided into two equal parts is $\binom{4n}{2n}$. The number of ways $2n$ boys can be divided evenly is $\binom{2n}{n}$ and the same holds for girls. Thus,

$$p_n = \frac{\binom{2n}{n} \binom{2n}{n}}{\binom{4n}{2n}} = \frac{[(2n)!]^4}{(n!)^4 (4n)!} \quad (3.30)$$

which, based on (3.29), can be approximated by

$$\frac{\left[\sqrt{2\pi}(2n)^{2n+\frac{1}{2}}e^{-2n}\right]^4}{\left[\sqrt{2\pi}n^{n+\frac{1}{2}}e^{-n}\right]^4 \left[\sqrt{2\pi}(4n)^{4n+\frac{1}{2}}e^{-4n}\right]} = \sqrt{\frac{2}{n\pi}}.$$

For example, in the case of 16 boys and 16 girls ($n = 8$), the approximate chances of dividing both sexes evenly are about $\sqrt{1/(4\pi)} = 0.2821$. The exact value is 0.2756.

PROBLEMS

3.4.1 Show that if $a \leq b \leq c \leq n$, then

$$\binom{n}{c} \binom{c}{b} \binom{b}{a} = \binom{n}{a} \binom{n-a}{b-a} \binom{n-b}{c-b}.$$

(i) Use the definition of binomial coefficients as ratios of the factorials. **(ii)** Use directly the interpretation of the binomial coefficients as the number of subsets of a given size. **(iii)** How many ways can one choose an a -element subset from a b -element subset from a c -element subset from a d -element subset from a n element set? (where $a \leq b \leq c \leq d \leq n$).

3.4.2 Find the coefficient of the term $x^4 y^5 z^3$ in the expansion of $(x - 2y + 3z)^{12}$.

3.4.3 Use the argument analogous to that in Theorem 3.3.2 to show that if $i \geq 1, j \geq 1$, and $k \geq 1$, then

$$\binom{n+1}{i, j, k} = \binom{n}{i-1, j, k} + \binom{n}{i, j-1, k} + \binom{n}{i, j, k-1}.$$

3.4.4 Use Stirling's formula to approximate the number of ways: **(i)** A set of size $2n$ can be partitioned into two equal parts. **(ii)** A set of size $3n$ can be partitioned into three equal parts.

3.4.5 Approximate the probability that every state will be represented in the committee in Problem 3.3.1 (i).